

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)
16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)

- 39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
- 40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Multi-Access Edge Computing: A Survey on Security, Dependability, and Performance

The report presents a survey of **Multi-Access Edge Computing (MEC) in 5G networks**, focusing on three important aspects: **security, dependability, and performance**. MEC brings computing and storage resources closer to end users, thereby reducing latency, network congestion, and dependence on centralized cloud services. It is particularly useful for applications such as **IoT, autonomous vehicles, industrial automation, and Ultra-Reliable Low-Latency Communication (URLLC)**.

The report explains the **ETSI MEC architecture**, which consists of MEC Hosts, virtualization infrastructure, MEC platforms, and management components such as the **MEC Orchestrator (MEO), MEC Platform Manager (MEPM), and Virtual Infrastructure Manager (VIM)**. MEC can be integrated with **5G networks, Network Function Virtualization (NFV), Software-Defined Networking (SDN), and network slicing** to provide flexible and efficient services.

Security is a major challenge because MEC nodes are distributed and may be deployed in locations with limited physical protection. The report discusses threats including eavesdropping, Man-in-the-Middle attacks, DoS/DDoS attacks, privacy violations, virtualization attacks, and unauthorized access. Encryption, authentication, access control, intrusion detection, secure communication, and AI/ML-based security mechanisms can help address these threats.

Dependability ensures that MEC services remain reliable and trustworthy. Important characteristics include availability, reliability, survivability, safety, and maintainability. Fault tolerance, redundancy, service migration, resource replication, and effective failure recovery are required to maintain service continuity, particularly in mobile and mission-critical applications.

Performance mainly concerns latency, throughput, jitter, packet loss, energy efficiency, and Quality of Service (QoS). Efficient resource allocation, MEC server placement, caching, congestion management, and service migration can improve performance. However, virtualization and security mechanisms may introduce additional overhead.

A key finding of the report is that **security, dependability, and performance are strongly interconnected**. Improving one aspect can negatively affect another; therefore, they must be considered jointly rather than independently. The report identifies **AI and machine learning, intelligent orchestration, MEC federation, and integrated monitoring** as important directions for future MEC and 6G networks.

In conclusion, MEC is a crucial technology for realizing the low-latency, reliable, and intelligent services expected from 5G and future 6G networks. Developing **secure, dependable, and high-performance MEC systems through integrated optimization and intelligent management** is essential for their successful deployment.

Reference:

G. Nencioni, R. G. Garroppo, and R. F. Olimid, "5G Multi-Access Edge Computing: A Survey on Security, Dependability, and Performance," *IEEE Access*, vol. 11, pp. 63496–63527, 2023



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5G Multi-Access Edge Computing: A Survey on Security, Dependability, and Performance

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ABSTRACT The Fifth Generation (5G) of mobile networks offers new and advanced services with stricter requirements. Multi-access Edge Computing (MEC) is a key technology that enables these new services by deploying multiple devices with computing and storage capabilities at the edge of the network, close to end-users. MEC enhances network efficiency by reducing latency, enabling real-time awareness of the local environment, allowing cloud offloading, and reducing traffic congestion. New mission-critical applications require high security and dependability, which are rarely addressed alongside performance. This survey paper fills this gap by presenting 5G MEC's three aspects: security, dependability, and performance. The paper provides an overview of MEC, introduces taxonomy, state-of-the-art, and challenges related to each aspect. Finally, the paper presents the challenges of jointly addressing these three aspects.

INDEX TERMS 5G, MEC, security, dependability, performance.